FINAL TESTING EVIDENCE -GROUP 2-

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1. Testing Overview

The final testing phase was conducted to verify that the Lifestyle Guru prototype performs according to the implemented project requirements and final approved scope. Testing focused on the major user workflows, including account access, profile onboarding and management, BMI and BMR calculations, nutrition goal settings, Dashboard progress tracking, Food Library functions, Favorites, Most Common Foods, meal logging, meal editing and deletion, calendar navigation, and date-based nutrition history.

The final prototype was tested using the team's defined set of 45 test cases (TC-01 through TC-45). Each test case was reviewed against the completed application and assigned a final status of Pass, Pass with Limitation, or Omitted. Tests marked Pass confirmed that the implemented feature behaved as expected. Tests marked Pass with Limitation identified functionality that worked but differed from the originally planned behavior. Tests marked Omitted represented features that were intentionally removed or revised as part of the final project scope.

Testing was performed on the final integrated version of the application using the Flask web application, SQLite database, and browser-based interface. After defects were identified and corrected, affected features were re-tested to confirm that the fixes worked and did not interfere with other major functions. Screenshot evidence was also collected to document the final behavior of the system and support the recorded test results.

2. Final Test Case Results

The following table presents the final results for test cases TC-01 through TC-45. Each test case includes the requirement or feature being tested, test preconditions, action/input, expected result, actual result, and final testing status. These results reflect the final tested state of the Lifestyle Guru prototype.

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-01	Welcome Screen	Application is available	Open Lifestyle Guru	"Welcome to Lifestyle Guru" is displayed	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-02	Login Screen	Welcome screen displayed	Continue to account-access screen	Lifestyle Guru name, tagline, Sign Up, and Login controls appear	As expected	Pass
TC-03	Account Creation	User has no account	Attempt account creation using supported sign-up method	Selected sign-up method opens/functions as designed	Omit supported sign-up, instead it leads to onboarding process	Omit
TC-04	Returning User Login	Valid account exists	Enter valid login credentials	Login succeeds and user proceeds to profile selection/application access	Omit supported sign-up, only sign in with test account which is a username ends with @gmail.com and random password	Omit
TC-05	Dashboard Components	Valid profile selected	Open Dashboard	Profile, Calendar, calorie totals/goals, and protein/fat/carb totals/goals are shown	As expected	Pass
TC-06	View Profile Information	Profile exists	Open profile	Stored height, weight, sex/gender, activity level and	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
				calculated BMI/BMR are displayed correctly		
TC-07	Height Conversion	Profile height entered in inches	Enter known inch value	Internal centimeter value matches required conversion	Only as new user when creating account (on boarding), not in profile adjustment	Pass with limitation
TC-08	Weight Conversion	Profile weight entered in pounds	Enter known pound value	Internal kilogram value matches required conversion	Only as new user when creating account (on boarding), not in profile adjustment	Pass with limitation
TC-09	BMI/BMR Calculation	Valid profile values exist	Submit known height, weight, sex/gender, and activity values	BMI and BMR match expected test values	After clicking save, not at instance changing other values	Pass with limitation
TC-10	Select Profile Screen	Login succeeds	Complete login	Select Profile screen is displayed	Omit multiple profiles on 1 account	Omit
TC-11	Add Profile Access	No profile exists	Select Add Profile	Profile-creation interface opens	Omit multiple profiles on 1 account	Omit

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-12	Create Profile	User is in Add Profile workflow	Enter valid required profile information and save	Profile is accepted and stored	Omit	Omit
TC-13	Internal Measurement Storage	Profile values entered	Save profile	Height is stored in centimeters and weight in kilograms	As expected	Pass
TC-14	Edit Profile	Existing profile exists	Modify profile information and save	New values replace previous stored values	As expected	Pass
TC-15	Delete Profile	Existing profile exists	Delete selected profile	Selected profile is removed	Omit	Omit
TC-16	Create Nutrition Goals	Profile exists	Enter calorie and macro goals	Goal information is accepted and stored	As expected	Pass
TC-17	Set Macro Percentages	Goal screen open	Adjust protein, fat, and carbohydrate percentages	User can modify all three percentage values	As expected	Pass
TC-18	Macro 100% Rule	Goal screen open	Change one macro percentage	Remaining values adjust so combined total remains 100%	A pop-up notice of “Protein + Carbs + Fats must total 100%”, instead of auto-adjust	Pass with limitation

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-19	Macro Allocation Calculation	Valid calorie goal and percentages exist	Save goal values	Calculated macro calorie allocation matches selected percentages	As expected	Pass
TC-20	Edit Goals	Existing goals exist	Modify calorie/macro targets	Updated values are saved and displayed	As expected	Pass
TC-21	Delete Goals	Existing goals exist	Delete saved goals	Selected goal information is removed	No delete option, but user can reset to 2000 kcal, 25% protein, 50% carb, 25% fat	Pass with limitation
TC-22	Log Meal to Current Date	Profile exists	Log a meal	Meal record is stored under the current date	As expected	Pass
TC-23	Meal Entry Options	Log Meal screen open	Begin meal logging	“Select Item from Food Library” and “Add New Item” are both available	As expected	Pass
TC-24	Meal Calorie Calculation	Valid food values entered	Save meal	Calculated meal calorie information matches expected values	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-25	Edit Logged Meal	Meal exists for selected date	Change meal information and save	Meal is updated and nutrition totals are recalculated	Only works if “edit” is being done inside “Your Daily Menu”; doesn’t work if the same made inside Library	Pass with limitation
TC-26	Delete Logged Meal	Existing meal record	Delete selected meal	Meal is removed and totals update accordingly	As expected	Pass
TC-27	Add Food Library Item	Food Library accessible	Select Add New Item	Custom food-entry form opens	As expected	Pass
TC-28	Required Food Fields	Add-food form open	Enter name, meal type, calories, protein, carbs, and fats	All required values can be entered and saved	As expected	Pass
TC-29	Meal Categories	Add/log meal screen open	View meal-type options	Breakfast, Lunch, Dinner, and Snacks are available	As expected	Pass
TC-30	Calendar Navigation	Dashboard/calendar accessible	Navigate through dates	Current, past, and future dates can be viewed and selected	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-31	Date Nutrition Totals	Selected date has meal data	Select that date	Correct total calories and macronutrients are displayed	As expected	Pass
TC-32	Four-Tier Progress Display	Calendar/date view open	View nutrition progress	Calories, Fats, Protein, and Carbs progress categories appear	As expected	Pass
TC-33	Goal Progress Visual Change	Nutrition goal is reached	Log enough nutrition to reach target	Progress display changes according to planned goal-reached appearance	As expected	Pass
TC-34	Selectable Calendar Days	Calendar displayed	Click a day	Selected date becomes active/open	As expected	Pass
TC-35	Open Day Meal History	Selected date contains meals	Open selected day	Stored meals for that date are displayed	As expected	Pass
TC-36	Meal Category Organization	Selected date contains multiple meals	View Daily Menu	Meals appear under Breakfast, Lunch, Dinner, or Snacks correctly	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-37	Food Library Persistence	Food item has been saved	Leave and return to Food Library	Saved item remains available for reuse	As expected	Pass
TC-38	Food Library Ordering	Multiple foods exist	Open Food Library	Items appear in the approved alphabetical ordering	As expected	Pass
TC-39	Favorite Food	Saved food exists	Mark food as favorite	Favorite status is stored/applied	As expected	Pass
TC-40	Favorites List	One or more foods are favorited	Open Favorites	Favorited food items appear in the Favorites List	As expected	Pass
TC-41	Most Common Foods	Foods have been logged at different frequencies	Open Most Common Foods	Frequently logged items are identified based on usage	As expected	Pass
TC-42	Most Common Foods in Meal Workflow	Usage history exists	Choose Select Item from Food Library while logging a meal	Most Common Food Item list is accessible	As expected	Pass

Test #	Requirement / Feature	Preconditions	Action / Input	Expected Result	Actual Result	Pass/ Fail
TC-43	Edit Food Item	Saved Food Library item exists	Edit food information and save	Existing values load and revised values replace them	As expected	Pass
TC-44	Delete Food Item	Saved Food Library item exists	Attempt permanent deletion	Selected item is removed after confirmation	As expected	Pass
TC-45	Delete Confirmation	Saved Food Library item exists	Select Delete	“Are you sure?” prompt with Yes/No appears; No cancels and Yes confirms deletion	As expected	Pass

3. Testing Evidence and Screenshots

The following screenshots provide visual evidence of the final testing results for the Lifestyle Guru prototype. Each figure is labeled with the test case or test cases it supports. The screenshots were selected to demonstrate the major implemented workflows, including account access, profile management, nutrition goals, Dashboard progress, Food Library functions, meal logging, and date-based nutrition history.

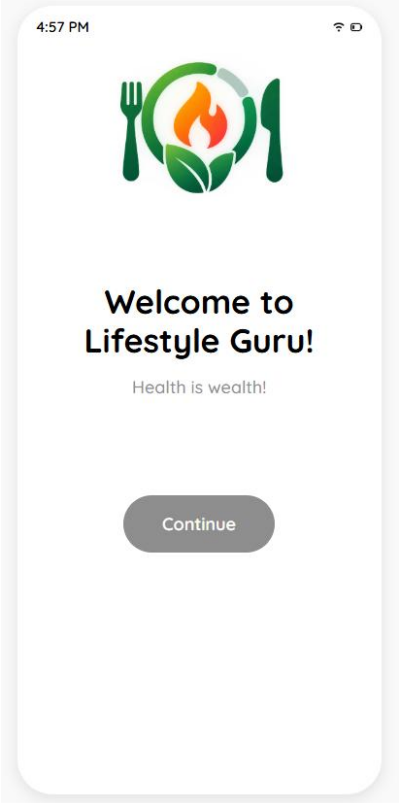


Figure 1 – TC-01: Welcome Screen
Evidence: The Lifestyle Guru welcome screen is displayed correctly.

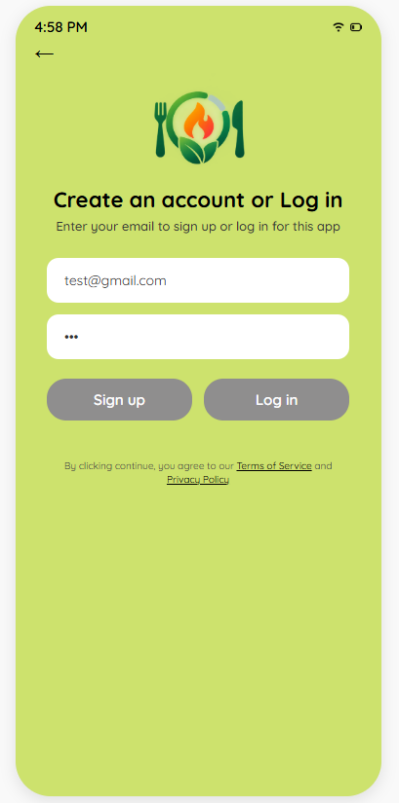


Figure 2 – TC-02: Login Screen
Evidence: The login screen displays the account-access controls used by the final prototype.

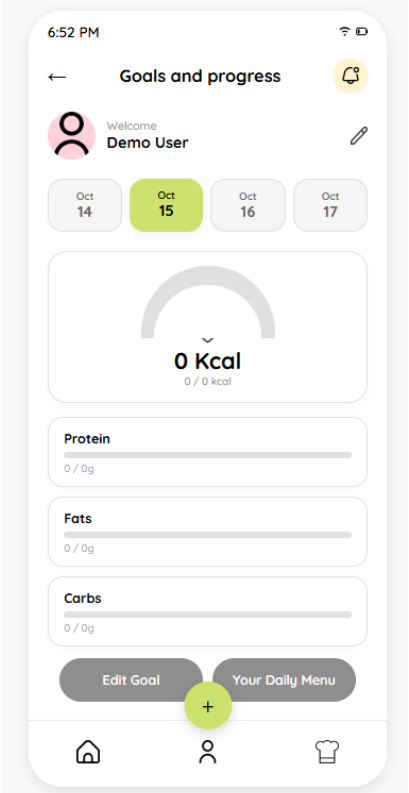


Figure 3 – TC-05: Dashboard Components
Evidence: The Dashboard displays the user profile summary, calorie information, macronutrient information, and navigation controls.

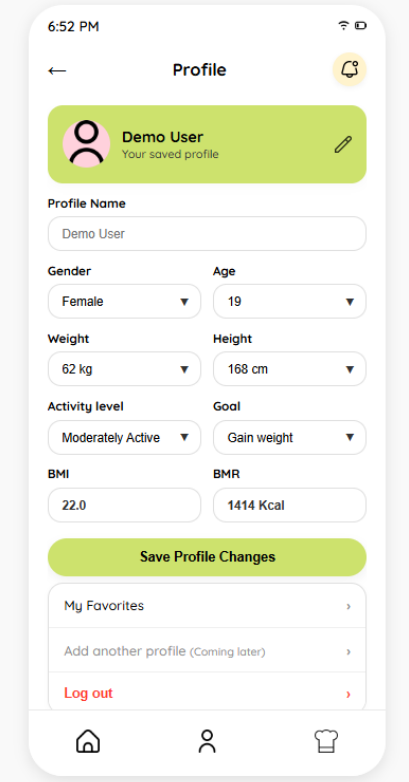


Figure 4 – TC-06 and TC-13: Profile Information and Internal Measurement Storage
Evidence: The profile screen displays stored profile information and the saved measurement values used by the application.

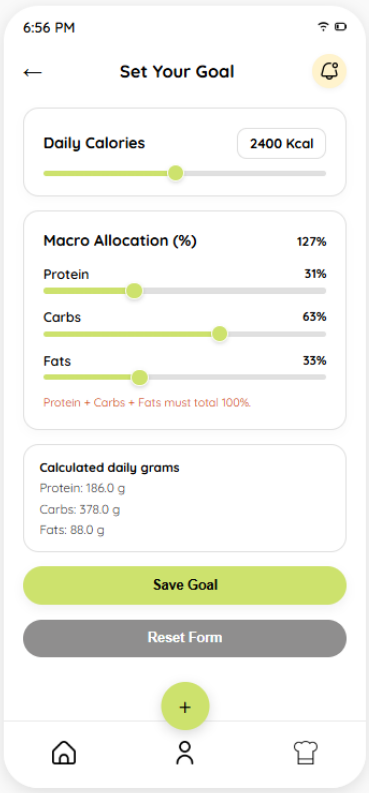


Figure 5 – TC-18: Macro 100% Validation Rule
Evidence: The nutrition-goal screen displays the validation behavior requiring protein, carbohydrates, and fats to total 100%.

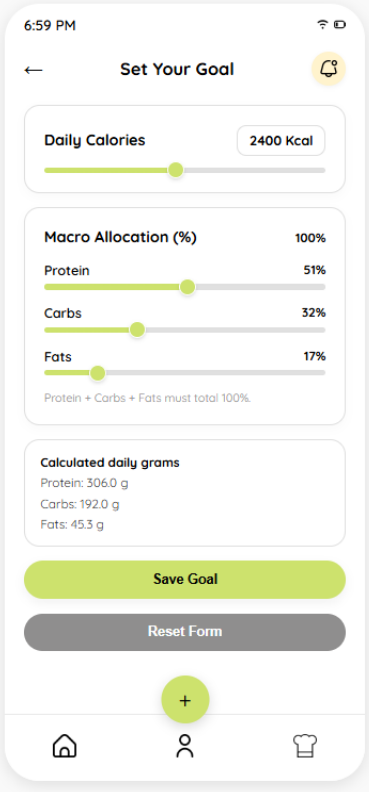


Figure 6 – TC-19: Macro Allocation Calculation
Evidence: The nutrition-goal screen shows calculated macronutrient allocations based on the saved calorie goal and percentage values.

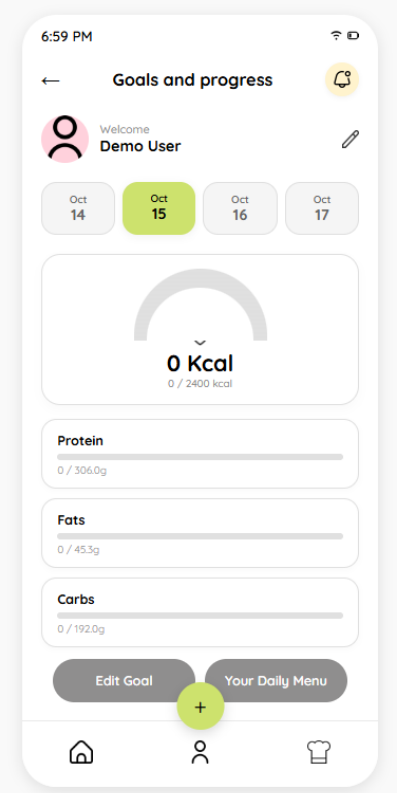


Figure 7 – TC-19 and TC-20:Macro Calculation and Goal Editing
Evidence: The updated nutrition goals are saved, and the calculated macronutrient values are displayed correctly.

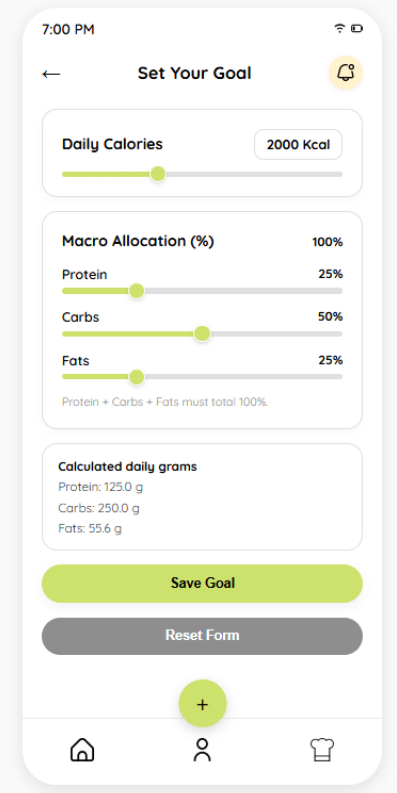


Figure 8 – TC-21: Nutrition Goal Reset
Evidence: The goal-management screen demonstrates the final reset behavior used instead of permanent goal deletion.

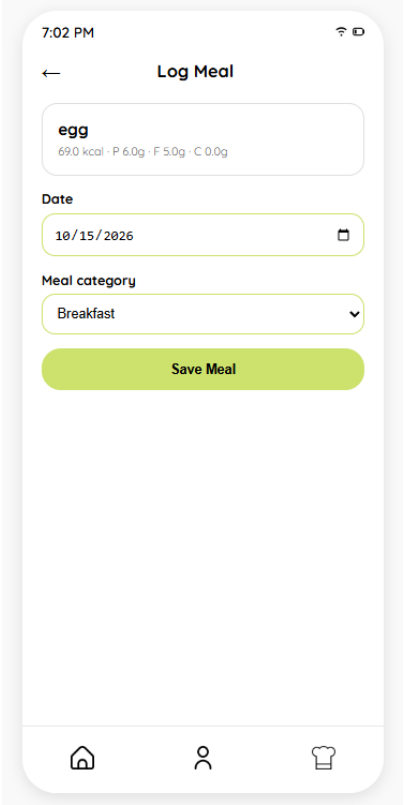


Figure 9 – TC-22: Log Meal to Current Date
Evidence: The meal-logging screen allows a meal to be entered for the selected date.

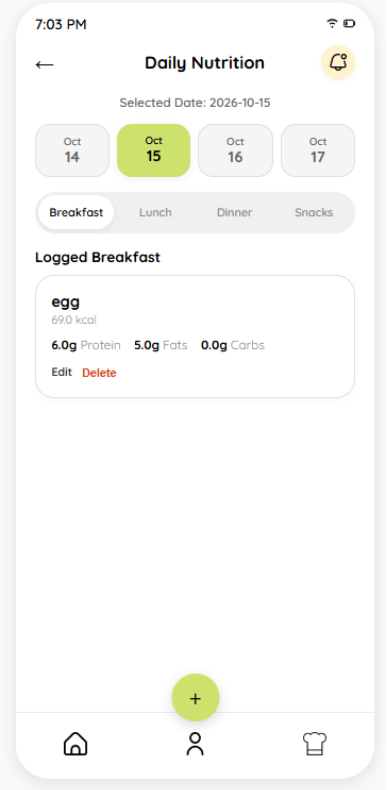


Figure 10 – TC-22: Logged Meal Display
Evidence: The newly logged meal appears in the Daily Menu for the selected date.

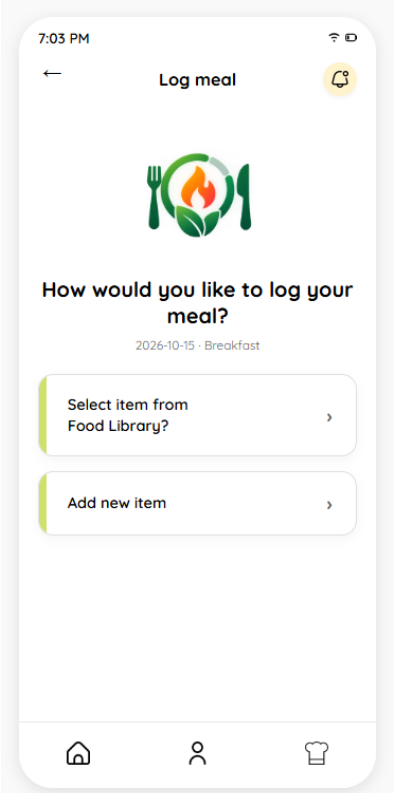


Figure 11 – TC-23: Meal Entry Options
Evidence: The Log Meal screen provides options to select an existing Food Library item or add a new item.

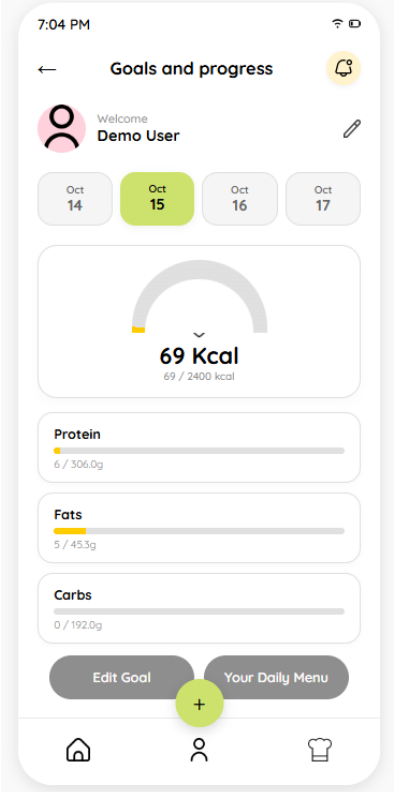


Figure 12 – TC-24: Meal Calorie Calculation
Evidence: The Dashboard reflects the calorie and nutrition totals after the meal is logged.

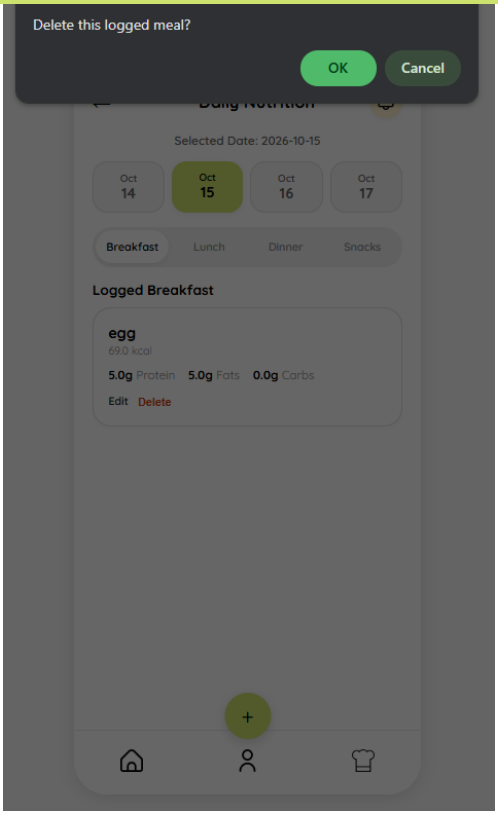


Figure 13 – TC-26 and TC-45: Delete Logged Meal and Confirmation
Evidence: The application displays a confirmation prompt before deleting a logged meal.

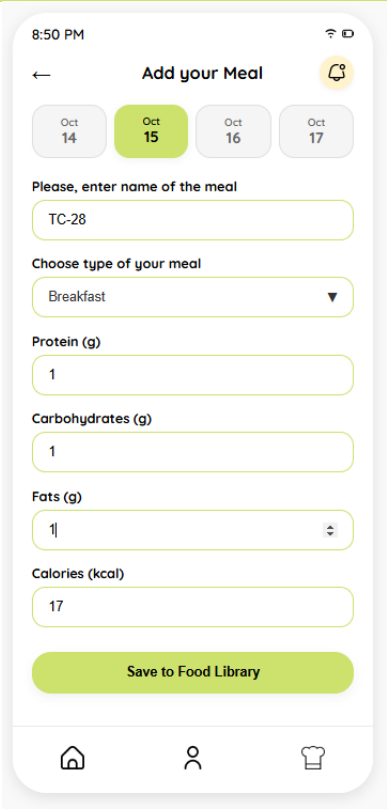


Figure 14 – TC-28: Required Food Fields
Evidence: The Add Food Item screen provides the required fields for entering and saving a custom food item.

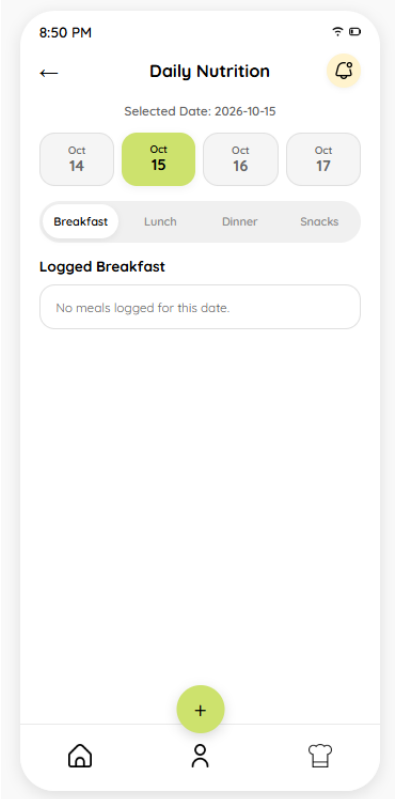


Figure 15 – TC-29: Meal Categories
Evidence: The application provides the supported meal categories, including Breakfast, Lunch, Dinner, and Snacks.

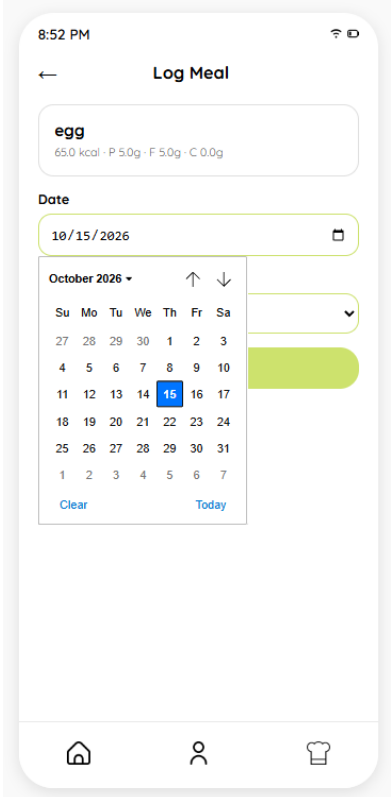


Figure 16 – TC-30: Calendar Navigation
Evidence: The calendar interface allows the user to navigate between and select different dates.

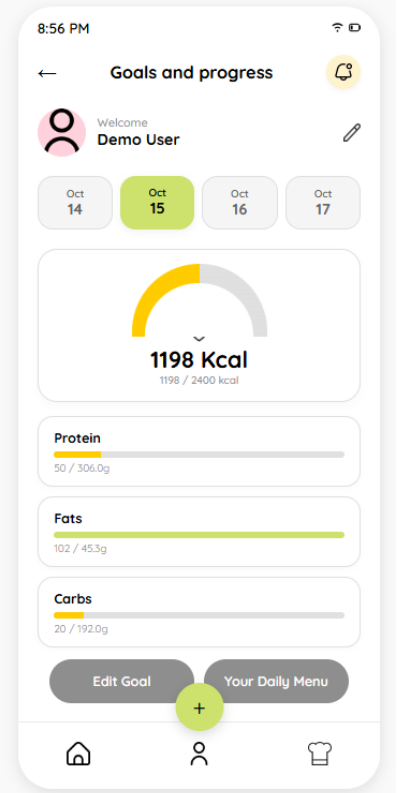


Figure 17 – TC-32 and TC-33: Nutrition Progress Display
Evidence: The Dashboard displays calorie and macronutrient progress and updates the visual indicators based on logged nutrition.

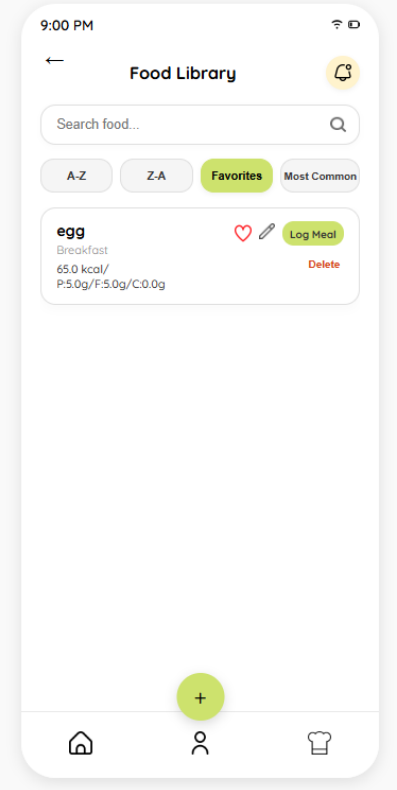


Figure 18 – TC-40: Favorites List
Evidence: The Favorites view displays food items that have been marked as favorites.

4. Pass with Limitation / Omitted Test Explanations

Several test cases were completed with limitations or were intentionally omitted because the final implementation differed from the originally planned design. These results do not indicate unexpected system failures; they document approved scope changes and differences between the original expected behavior and the completed prototype.

a) Pass with Limitation

i. **TC-07 – Height Conversion**
The height-conversion function works during the initial onboarding process. However, the same automatic conversion is not performed when the user later edits profile information. Because the required conversion is supported during profile creation but not throughout all profile-editing workflows, this test is recorded as Pass with Limitation.

ii. **TC-08 – Weight Conversion**
Weight conversion is supported during initial onboarding, where entered pound values are converted to the internal kilogram value used by the application. The conversion is not automatically repeated during later profile adjustments. Therefore, the test is recorded as Pass with Limitation.

iii. **TC-09 – BMI/BMR Calculation**
BMI and BMR calculations produce the expected values using the saved profile information. The limitation is that the displayed BMI and BMR values update after the user saves the profile changes rather than changing immediately while profile values are being edited.

iv. **TC-18 – Macro 100% Rule**
The final application validates that protein, carbohydrate, and fat percentages must total 100%. Instead of automatically adjusting the remaining macro percentages, the application displays a notice informing the user that the values must total 100%. The validation requirement is therefore enforced using a different behavior than originally expected.

v. **TC-21 – Delete Goals**
The final prototype does not permanently delete nutrition goals. Instead, the user can reset the goals to the default values of 2000 calories, 25% protein, 50% carbohydrates, and 25% fat. Because the final implementation provides reset behavior rather than deletion, this test is recorded as Pass with Limitation.

vi. **TC-25 – Edit Logged Meal**
Logged meals can be edited successfully through the Your Daily Menu workflow, and the nutrition totals are recalculated after the changes are saved. However, editing the related item in the Food Library does not change a meal that has already been logged. The test is therefore recorded as Pass with Limitation.

b) Omitted Tests

i. **TC-03 – Account Creation**
The originally planned separate sign-up/account-creation process was removed from the final project scope. The final prototype instead uses simplified account access that leads into the onboarding process when profile information has not yet been created. Therefore, TC-03 was Omitted.

ii. **TC-04 – Returning User Login**
The original test expected a returning user to proceed through the planned account and profile-selection workflow. The final prototype uses simplified login behavior rather than the originally designed returning-user/profile-selection process. Therefore, TC-04 was Omitted.

iii. **TC-10 – Select Profile Screen**
The final prototype supports one active profile rather than multiple profiles. Because there is no need for a profile-selection screen, this test was intentionally Omitted.

iv. **TC-11 – Add Profile Access**
Adding additional profiles to the same account is not included in the final project scope. The Add Profile workflow was therefore not implemented, and TC-11 was Omitted.

v. **TC-12 – Create Profile**
This test referred specifically to creating an additional profile through the original Add Profile workflow. Because the final prototype uses one active profile and does not support multiple-profile creation, TC-12 was Omitted.

vi. **TC-15 – Delete Profile**
Profile deletion was removed from the final scope along with the multiple-profile workflow. The completed prototype allows the active profile to be viewed and edited but does not provide a separate profile-deletion function. Therefore, TC-15 was Omitted.

Overall, the tests marked Pass with Limitation represent features that remain functional but behave differently from the original expected design. The Omitted tests represent functionality that was intentionally removed or revised as part of the final project scope rather than unresolved defects.

5. Regression and Final Validation

After the initial test execution, the team reviewed the results to identify defects, incomplete behavior, and differences between the original design and the final implementation. Features affected by corrections were re-tested to confirm that the changes worked as intended and did not introduce problems into related parts of the application.

One of the main issues identified during final testing involved the Dashboard progress display. TC-33 originally did not behave as expected because the visual progress indicator was not updating correctly when nutrition values approached or reached the configured goals. The issue was corrected during the final development phase, and the feature was re-tested. The final result for TC-33 was recorded as Pass, confirming that the progress display responded correctly after the fix.

Regression testing was also performed across the major workflows of the application, including profile management, nutrition goals, Food Library functions, meal logging, Dashboard totals, Favorites, Most Common Foods, calendar navigation, and nutrition history. This helped confirm that final changes did not break features that had already passed earlier testing.

As part of final validation, the completed project was also tested in a fresh Python environment. The final branch was placed in a new project folder, a new virtual environment was created, and the required dependencies were installed from requirements.txt. The application was then started using python app.py, and the Lifestyle Guru web application launched successfully in the browser. This confirmed that the project could be installed and run using the instructions included with the final submission.

The SQLite database was also cleaned before final packaging so that previous testing data would not appear when the instructor first runs the application. After cleanup, the application was tested again to confirm that a new user could proceed through the onboarding workflow and use the system normally.

The final regression and validation process confirmed that the major implemented features were functional, the corrected Dashboard behavior remained stable, and the final submission package could be installed and executed successfully in a clean Python environment.

6. Final Testing Summary

The final testing process evaluated 45 test cases (TC-01 through TC-45) covering the major functional areas of the Lifestyle Guru prototype. The results showed that most of the implemented features operated as expected, while a small number of tests documented known limitations or intentional scope changes.

Final Status	Number of Test Cases
Pass	33
Pass with Limitation	6

Omitted	6
Fail	0
Total	45

The six tests recorded as Pass with Limitation were TC-07, TC-08, TC-09, TC-18, TC-21, and TC-25. These features remained functional but differed in some way from the originally defined expected behavior. The six tests recorded as Omitted were TC-03, TC-04, TC-10, TC-11, TC-12, and TC-15. These tests corresponded primarily to account-creation or multiple-profile functions that were removed or revised as part of the final project scope.

No test cases remained in a failed state after final corrections and regression testing. The completed system successfully demonstrated the primary Lifestyle Guru workflow, including profile management, BMI/BMR calculations, nutrition goals, Dashboard progress tracking, Food Library management, meal logging, Favorites, Most Common Foods, calendar navigation, and date-based nutrition history.

Overall, the final testing results indicate that the Lifestyle Guru prototype satisfies the major functional requirements included in the final project scope. Known limitations and omitted features have been documented rather than treated as unresolved defects, and the application was successfully validated for final submission.